

第六章 储能元件

- 电容元件
- 电感元件

§ 6-1 电容元件

电容元件



钽电解电容



独石电容



贴片电容

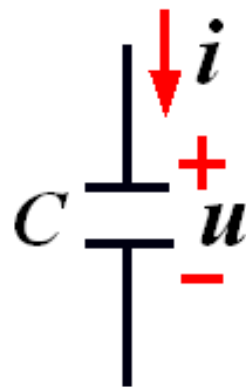
§ 6-1 电容元件

q - u 特性关系: $q = Cu$

$C = \frac{q}{u}$ —电容: 法拉

伏安特性: $i = \frac{dq}{dt} = C \frac{du}{dt}$

图
形
符
号



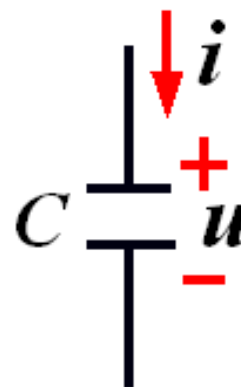
$$u = \frac{1}{C} \int_{-\infty}^t i dt = \frac{1}{C} \left[\int_{-\infty}^0 i dt + \int_0^t i dt \right] = u(0) + \frac{1}{C} \int_0^t i dt$$

§ 6-1 电容元件

电容元件的功率:

关联方向下: $P_C = ui = Cu \frac{du}{dt}$

图
形
符
号



$\frac{du}{dt} > 0$ $p_C > 0$ 吸收
电能

$\frac{du}{dt} < 0$ $p_C < 0$ 释放
电能

结
论

1. $\frac{du}{dt} = 0$ $i = 0$ 对于直流, 电容为开路元件。
2. 电容是储能 (记忆) 元件, 不消耗电能。

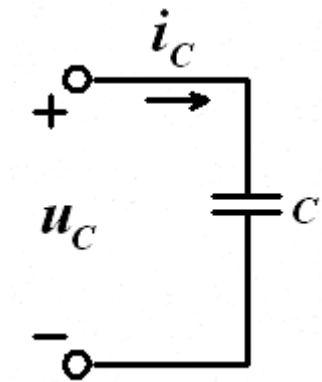
§ 6-1 电容元件

电容元件

$$u_C = \sqrt{2}U_C \sin(\omega t + \varphi_u)$$

$$i_C = C \frac{du_C}{dt}$$

$$i_C = \sqrt{2}\omega C U_C \sin\left(\omega t + \varphi_u + \frac{\pi}{2}\right) = \sqrt{2}I_C \sin(\omega t + \varphi_i)$$



相位差 $\varphi = \varphi_u - \varphi_i = -\frac{\pi}{2}$ 电压滞后电流 90°

§ 6-2 电感元件

电感元件



环形电感



电感线圈



贴片电感

§ 6-2 电感元件

电感元件伏安特性

韦安特性关系: $\psi = Li$

$L = \frac{\psi}{i}$ — 自感系数: 亨利



伏安特性:

$$u_L = L \frac{di}{dt}$$

$$i = \frac{1}{L} \int_{-\infty}^t u dt = \frac{1}{L} \left[\int_{-\infty}^0 u dt + \int_0^t u dt \right] = i(0) + \frac{1}{L} \int_0^t u dt$$

电感是动态元件, 也是记忆元件。

§ 6-2 电感元件

电感元件的功率:

关联方向: $P_L = u i = L i \frac{di}{dt}$



$$\frac{di}{dt} > 0 \quad p_L > 0 \quad \text{吸收电能}$$

$$\frac{di}{dt} < 0 \quad p_L < 0 \quad \text{释放电能}$$

结论

1. $\frac{di}{dt} = 0 \quad u = 0$ 对于直流, 电感为短路元件。
2. 电感是储能(记忆)元件, 不消耗电能。

§ 6-2 电感元件

电感元件

$$i_L = \sqrt{2}I_L \sin(\omega t + \varphi_i)$$

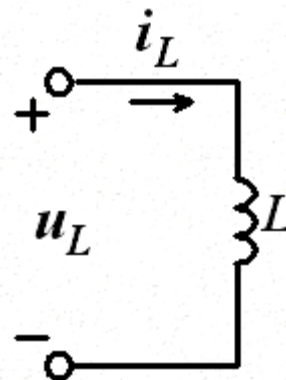
$$u_L = L \frac{di_L}{dt}$$

$$u_L = \sqrt{2}\omega LI_L \sin(\omega t + \varphi_i + \frac{\pi}{2})$$

$$= \sqrt{2}U_L \sin(\omega t + \varphi_u)$$

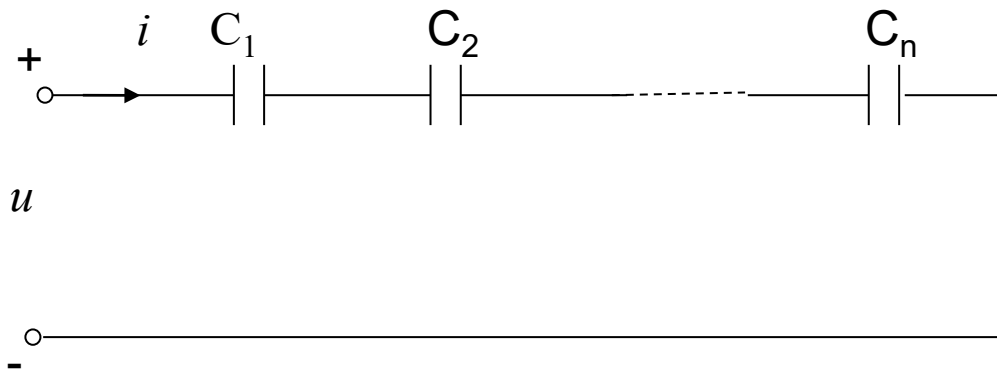
$$\text{相位差 } \varphi = \varphi_u - \varphi_i = \frac{\pi}{2}$$

电压超前电流 90°

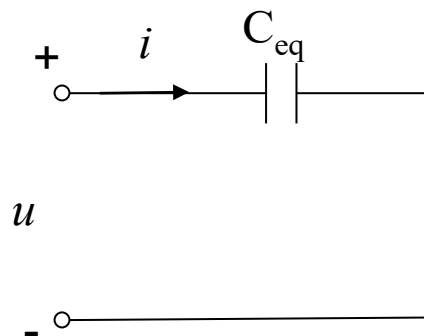


§ 6-3 电容、电感元件的串并联

■ 电容的串联

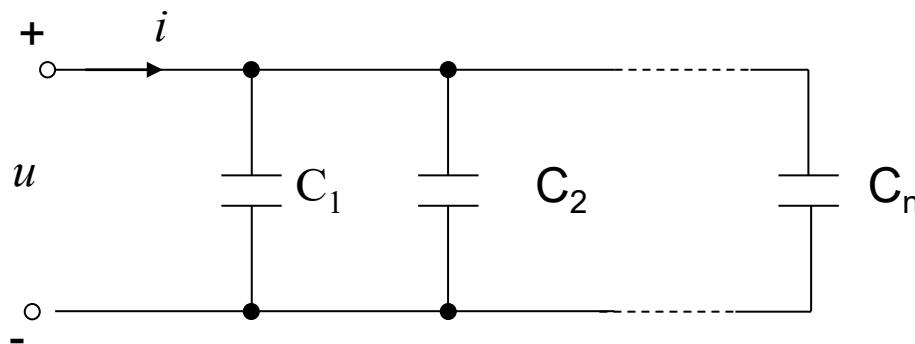


$$\frac{1}{C_{eq}} = \sum_{k=1}^n \frac{1}{C_k}$$

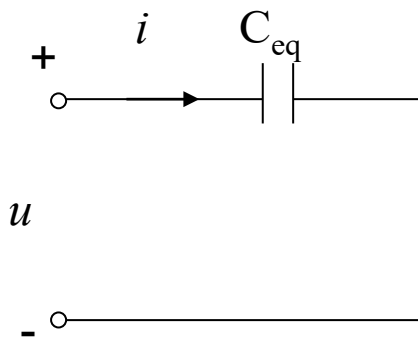


§ 6-3 电容、电感元件的串并联

■ 电容的并联

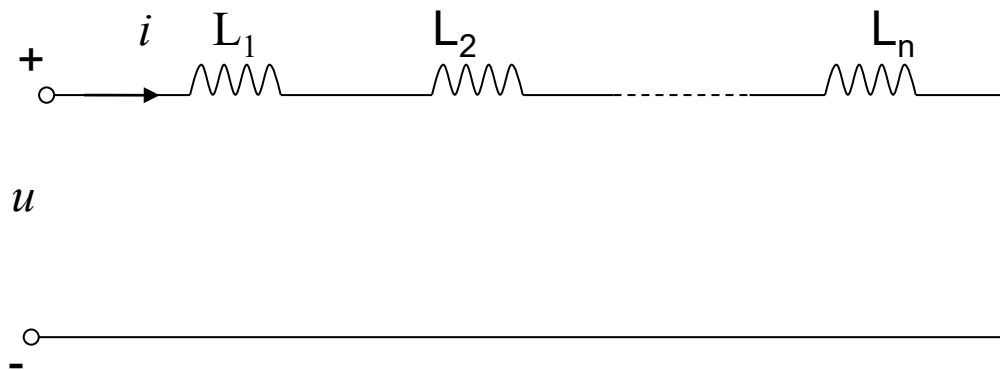


$$C_{eq} = \sum_{k=1}^n C_k$$

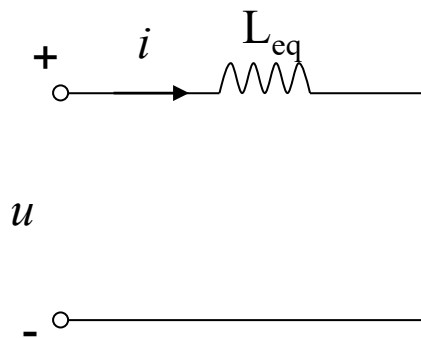


§ 6-3 电容、电感元件的串并联

■ 电感的串联

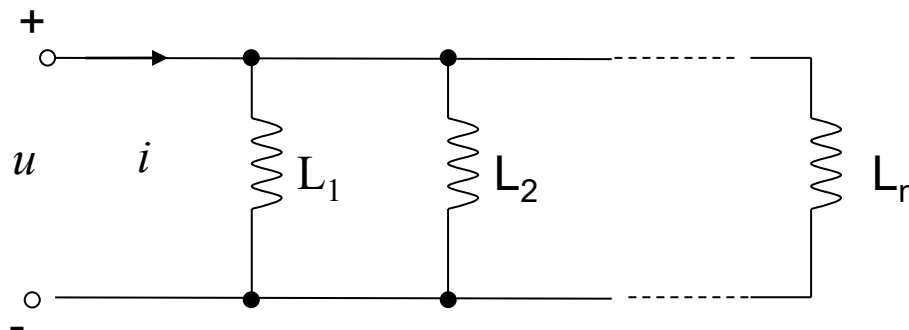


$$L_{eq} = \sum_{k=1}^n L_k$$

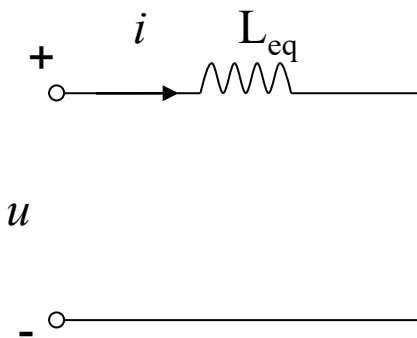


§ 6-3 电容、电感元件的串并联

■ 电感的并联



$$\frac{1}{L_{eq}} = \sum_{k=1}^n \frac{1}{L_k}$$



作业

■ P134

6-3

6-4

6-8

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